# HTE LITE

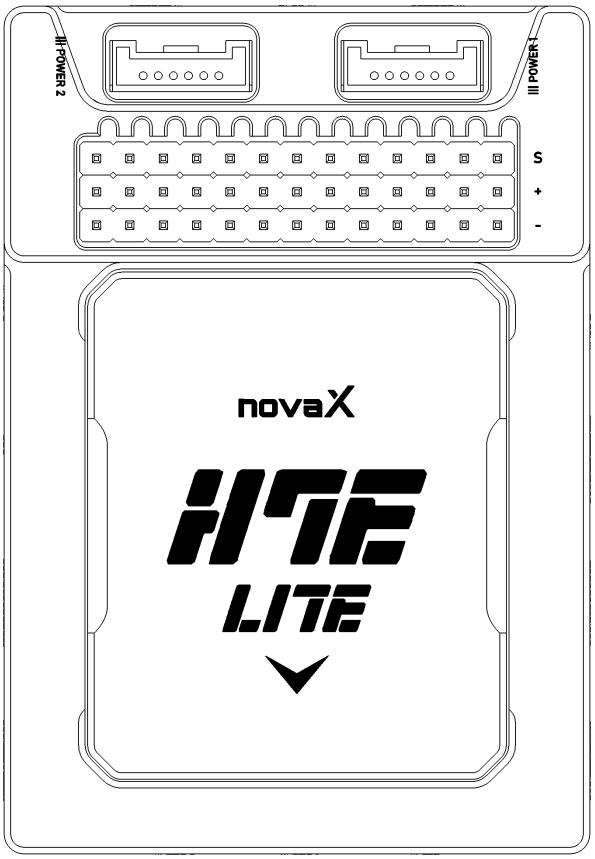
# User Guide



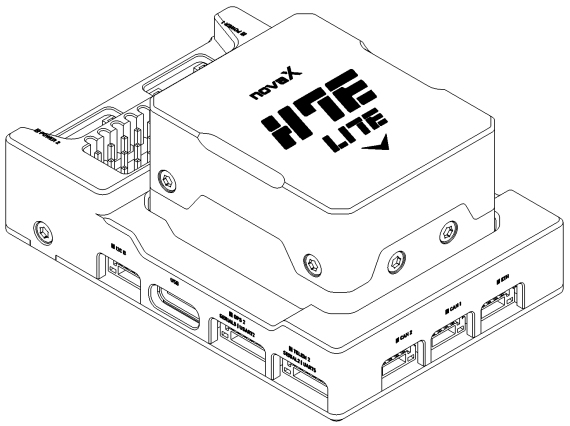

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# 1. Parts List

## · Main & Board




H7E Lite Autopilot



# 1. Parts List

## · Connector Type

| Port | Connector |
|---|---|
| POWER 1 · POWER 2 | Molex Micro-Lock Plus 6P (1.25 mm) |
| TELEM 1 · TELEM 2 · GPS 1 · GPS 2 | JST-GH 6P |
| SERIAL4 · SERIAL6 · I2C A · I2C B · CAN 1 · CAN 2 · ETHERNET | JST-GH 4P |
| RC IN | JST-GH 5P |
| DEBUG | JST-SH 6P |
| USB | USB Type-C |
| PWM OUT M1–M12 · SB (SBUS OUT) | 2.54 mm 3 × 13 pin header |

# 2. Specifications

## · Hardware Specifications (1)

| | |
|---|---|
| MCU | STM32H753, ARM Cortex-M7, 480 MHz |
| RAM / Flash | 1 MB / 2 MB |
| IMU | 2× ICM-42688-P + BMI088 |
| IMU Heater | Built-in heater on the IMU board, held at 45 °C |
| Magnetometer | RM3100 |
| Barometer | 2× ICP-20100 |
| Operating Voltage | 4.75 – 5.7 V (Rated 5 V) |
| USB Input | 4.75 – 5.25 V |
| Power Input | POWER 1 + POWER 2 (redundant, I2C power modules) |
| PWM Output | 12 FMU channels (DShot except M7–M8) + SBUS out |
| RC Input | S.Bus, PPM, DSM / Spektrum |
| RSSI Input | Analog |
| UART | 6 serial ports (TELEM 1 · 2, GPS 1 · 2, SERIAL4, SERIAL6) + debug console |
| GPS | GPS 1 + GPS 2 (6-pin, UART + I2C, left / right) |

# 2. Specifications

## · Hardware Specifications (2)

| | |
|---|---|
| I²C | 2 dedicated ports (I2C A, I2C B) + GPS 1 · 2 + power-module bus on POWER 1 · 2 |
| CAN | 2 Port |
| Ethernet | 100 Mbps x 1 Port |
| Size | 44.5 × 64.5 × 28.6 mm |
| Mount Hole | 16 × 30 mm / M3 (4 blind holes in the base, 3 mm thread) |
| Supported F/W | novaX ArduPilot |

# 2. Specifications

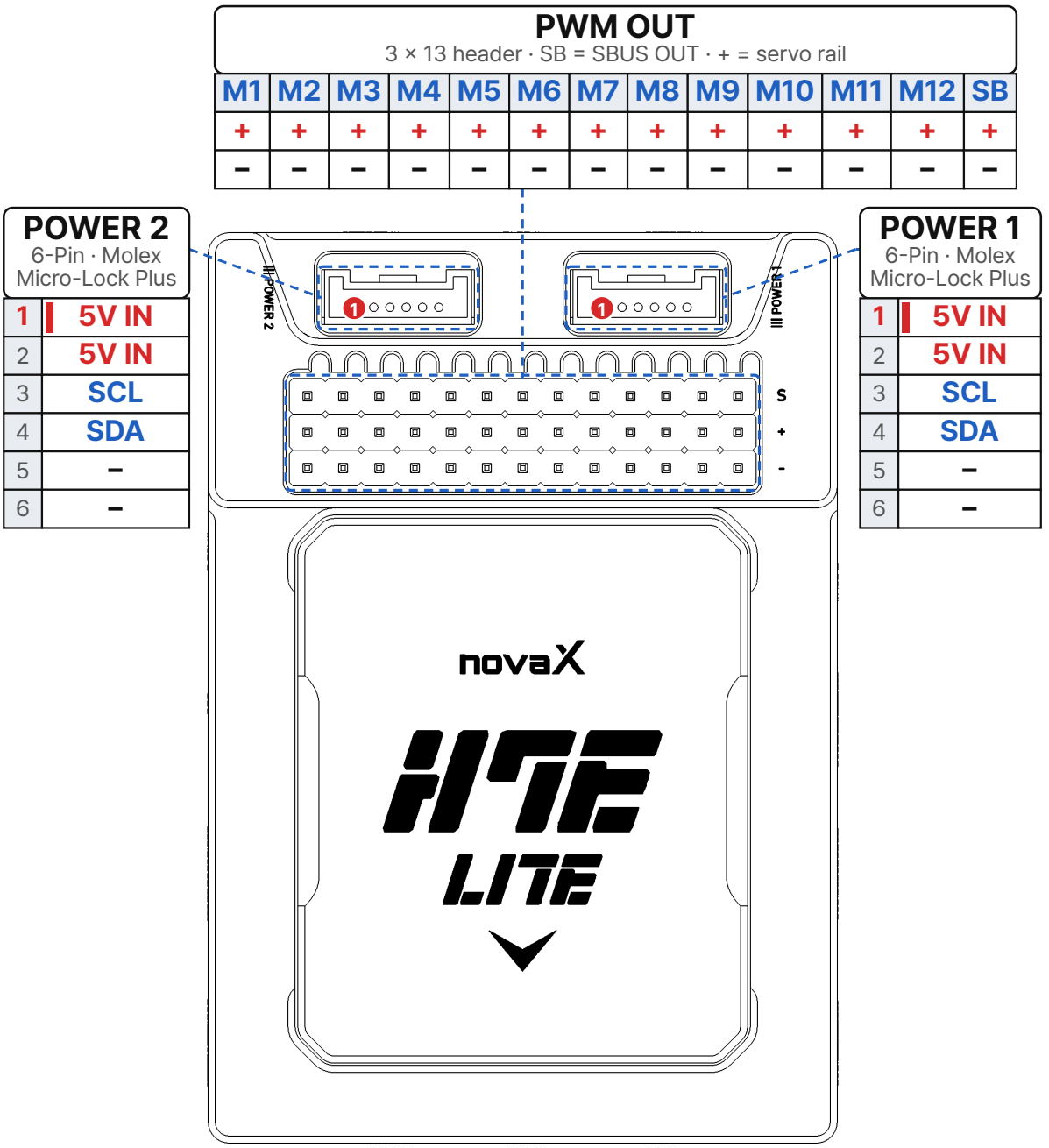
## · Interfaces

| | | | |
|---|---|---|---|
| POWER | 2 | ETHERNET | 1 |
| TELEM | 2 | DEBUG | 1 |
| GPS | 2 | USB | 1 |
| SERIAL | 2 | PWM Outputs | 12 |
| I2C | 2 | SBUS OUT | 1 |
| CAN | 2 | TF Card Slot | 1 |
| RC IN | 1 | | |
| Power Module | POWER 1 + POWER 2 (redundant, I2C power modules) | | |

# 3. Pinouts

## · Top

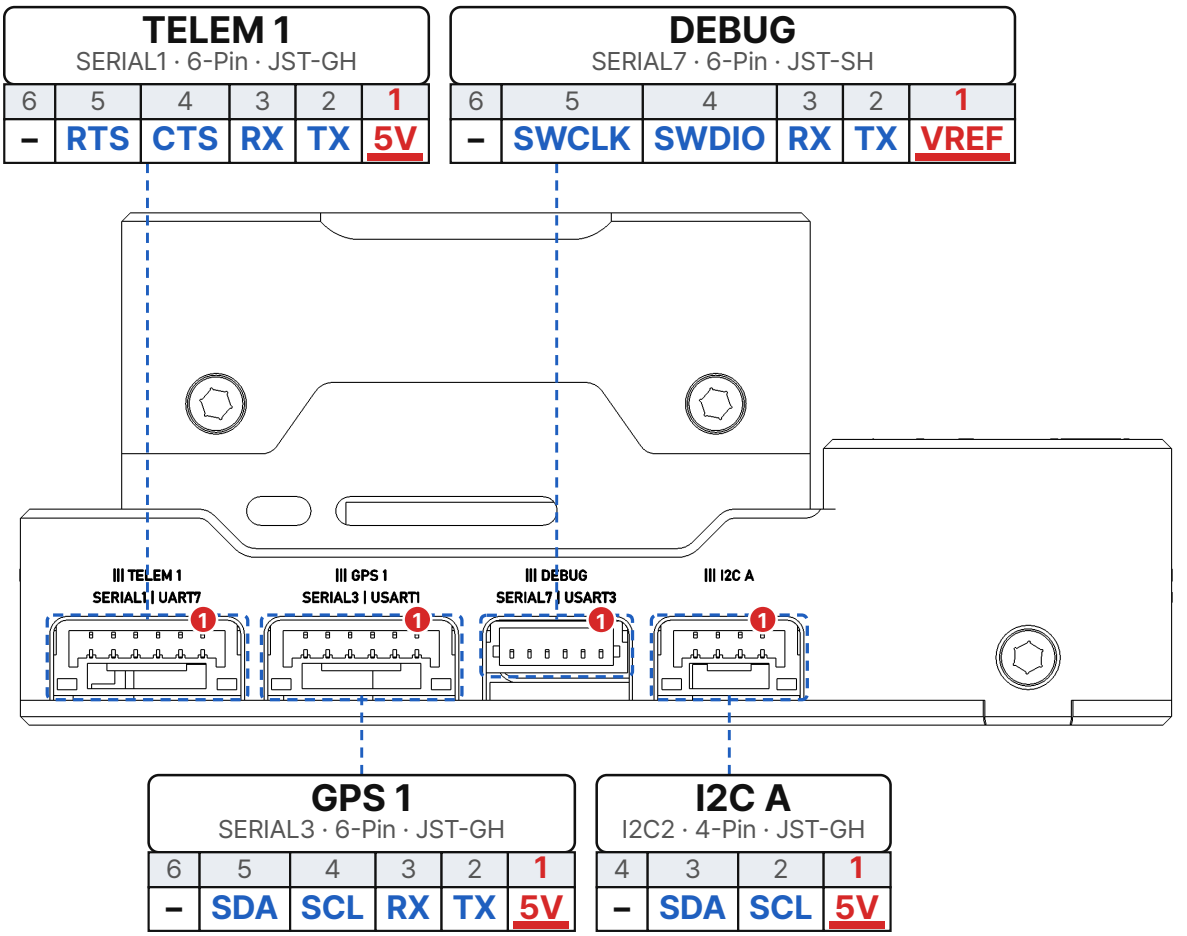
❶ = pin 1 **red = power** **– = GND** **blue = signal** · pins as seen in the picture



# 3. Pinouts

## · Left

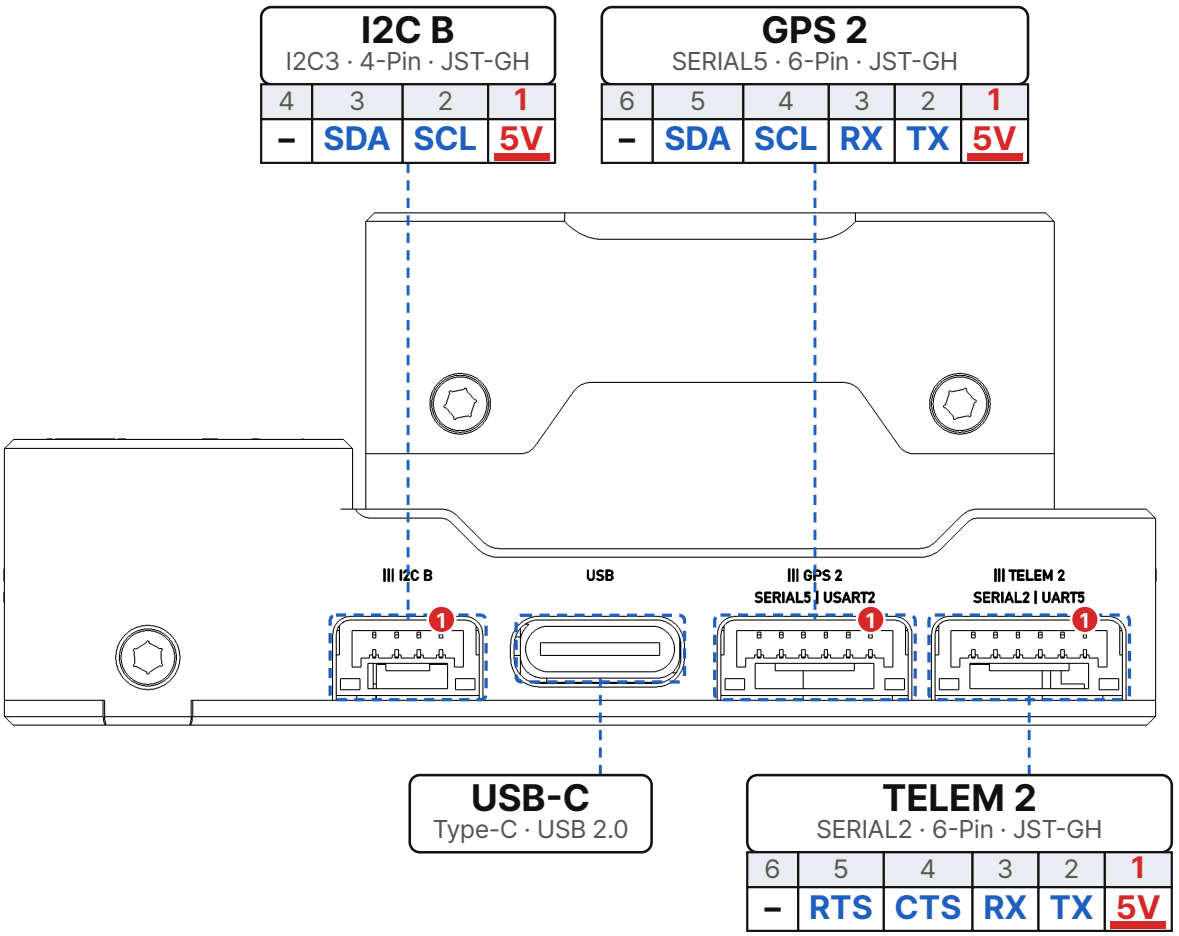
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# 3. Pinouts

## · Right

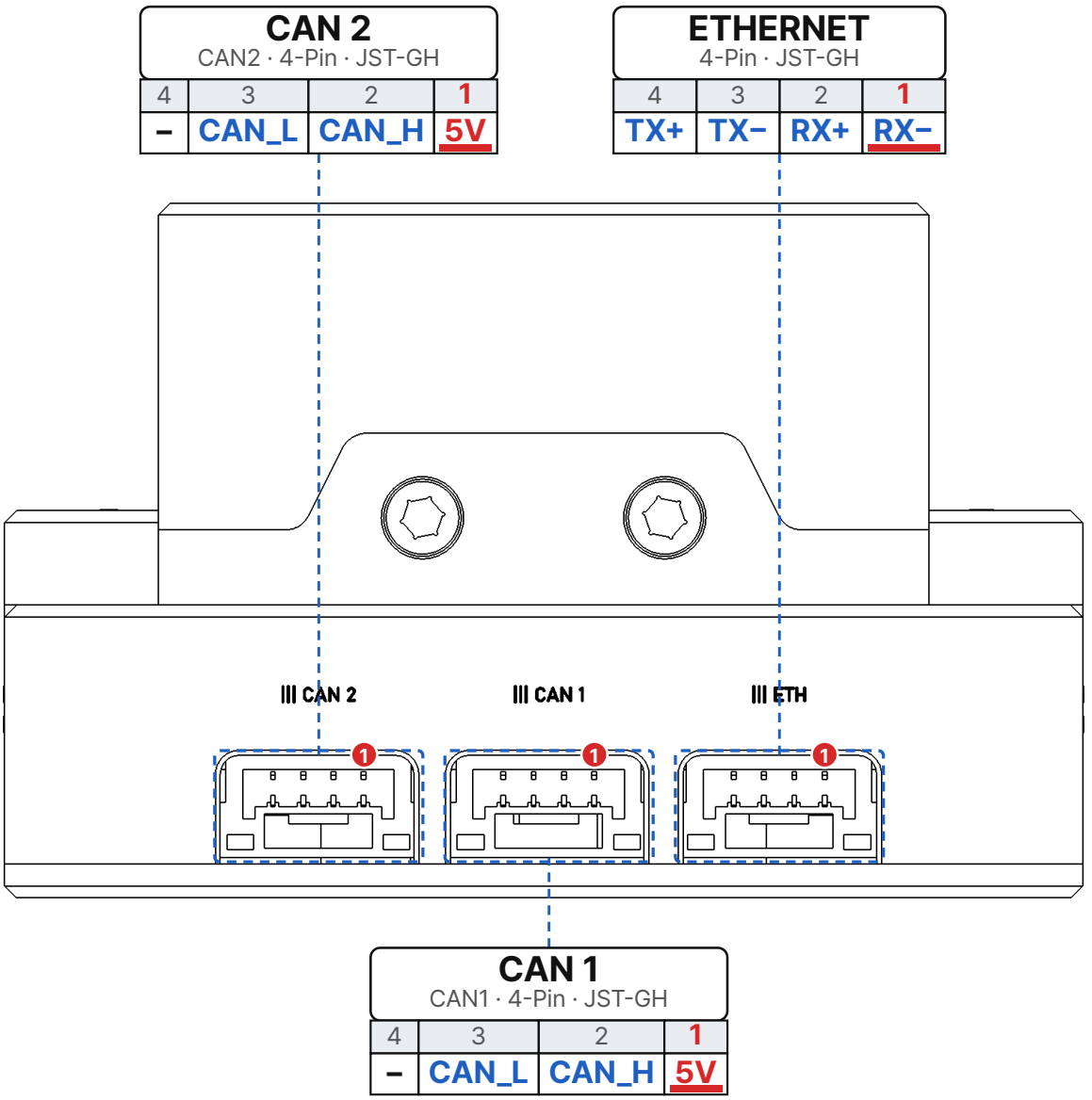
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# 3. Pinouts

## · Front

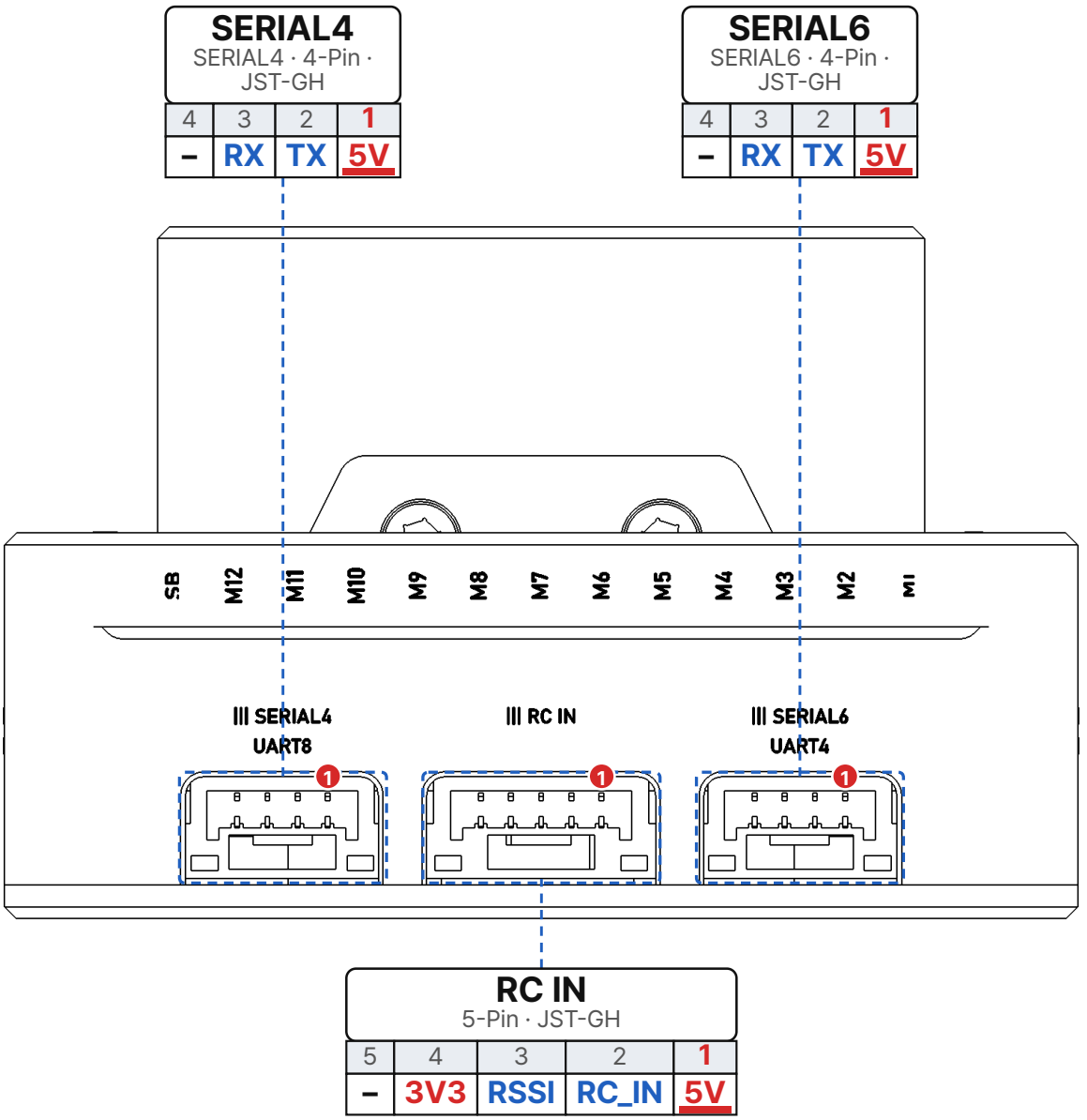
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# 3. Pinouts

## · Rear

1 = pin 1 **red = power** **– = GND** **blue = signal** · pins as seen in the picture



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## 3. Pinouts

### · Top

❶ = pin 1 **red = power** **– = GND** **blue = signal** · pins as seen in the picture

**PWM OUT**
3 × 13 header · SB = SBUS OUT · + = servo rail

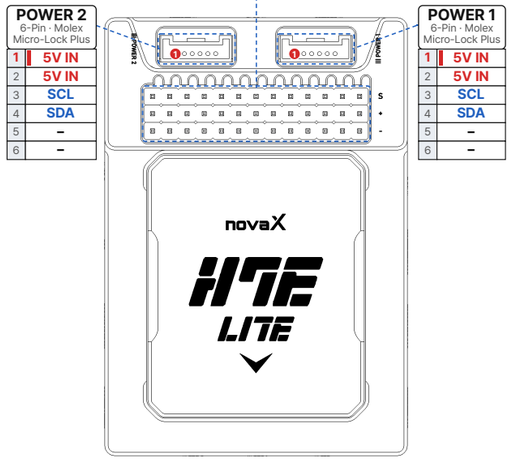
| M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 | M9 | M10 | M11 | M12 | SB |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| + | + | + | + | + | + | + | + | + | + | + | + | + |
| – | – | – | – | – | – | – | – | – | – | – | – | – |

**POWER 2**
6-Pin · Molex Micro-Lock Plus

| | |
|---|---|
| 1 | 5V IN |
| 2 | 5V IN |
| 3 | SCL |
| 4 | SDA |
| 5 | – |
| 6 | – |

**POWER 1**
6-Pin · Molex Micro-Lock Plus

| | |
|---|---|
| 1 | 5V IN |
| 2 | 5V IN |
| 3 | SCL |
| 4 | SDA |
| 5 | – |
| 6 | – |




## 3. Pinouts

### · Left

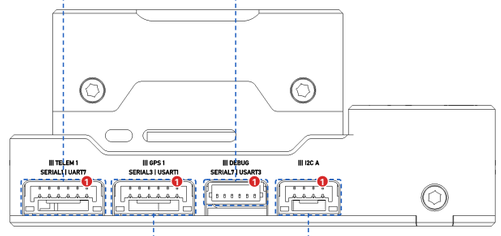
❶ = pin 1 **red = power** **– = GND** **blue = signal** · pins as seen in the picture

**TELEM 1**
SERIAL1 · 6-Pin · JST-GH

| 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| – | RTS | CTS | RX | TX | 5V |

**DEBUG**
SERIAL7 · 6-Pin · JST-SH

| 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| – | SWCLK | SWDIO | RX | TX | VREF |




**GPS 1**
SERIAL3 · 6-Pin · JST-GH

| 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| – | SDA | SCL | RX | TX | 5V |

**I2C A**
I2C2 · 4-Pin · JST-GH

| 4 | 3 | 2 | 1 |
|---|---|---|---|
| – | SDA | SCL | 5V |

## 3. Pinouts

### · Right

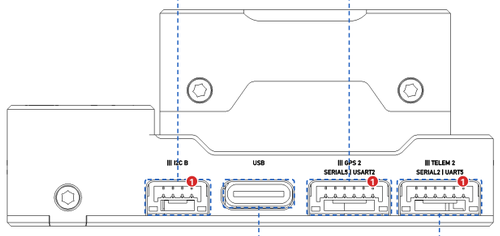
❶ = pin 1 **red = power** **– = GND** **blue = signal** · pins as seen in the picture

**I2C B**
I2C3 · 4-Pin · JST-GH

| 4 | 3 | 2 | 1 |
|---|---|---|---|
| – | SDA | SCL | 5V |

**GPS 2**
SERIAL5 · 6-Pin · JST-GH

| 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| – | SDA | SCL | RX | TX | 5V |




**USB-C**
Type-C · USB 2.0

**TELEM 2**
SERIAL2 · 6-Pin · JST-GH

| 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| – | RTS | CTS | RX | TX | 5V |

## 3. Pinouts

### · Front

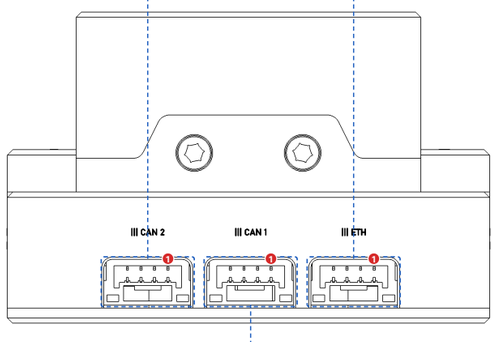
❶ = pin 1 **red = power** **– = GND** **blue = signal** · pins as seen in the picture

**CAN 2**
CAN2 · 4-Pin · JST-GH

| 4 | 3 | 2 | 1 |
|---|---|---|---|
| – | CAN_L | CAN_H | 5V |

**ETHERNET**
4-Pin · JST-GH

| 4 | 3 | 2 | 1 |
|---|---|---|---|
| TX+ | TX– | RX+ | RX– |




**CAN 1**
CAN1 · 4-Pin · JST-GH

| 4 | 3 | 2 | 1 |
|---|---|---|---|
| – | CAN_L | CAN_H | 5V |

## 3. Pinouts

### · Rear

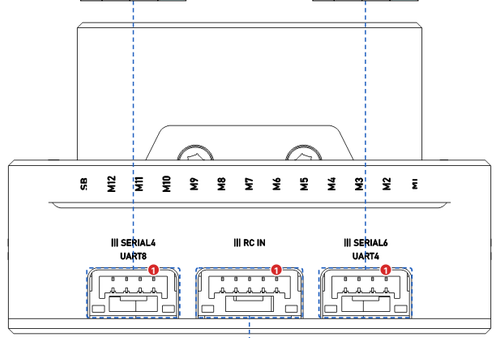
❶ = pin 1 **red = power** **– = GND** **blue = signal** · pins as seen in the picture

**SERIAL4**
SERIAL4 · 4-Pin · JST-GH

| 4 | 3 | 2 | 1 |
|---|---|---|---|
| – | RX | TX | 5V |

**SERIAL6**
SERIAL6 · 4-Pin · JST-GH

| 4 | 3 | 2 | 1 |
|---|---|---|---|
| – | RX | TX | 5V |




**RC IN**
5-Pin · JST-GH

| 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|
| – | 3V3 | RSSI | RC_IN | 5V |